Grid-Forming PMSG Wind Turbine with Rotor Kinetic-Energy Reserve

Step 1 Progress Report, Wind Electrical Systems

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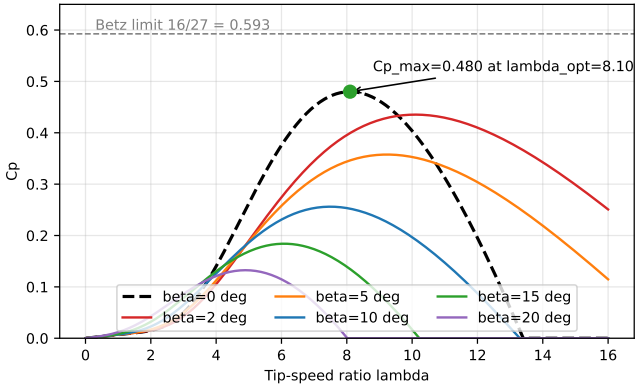
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12 October 2026

Problem and India context

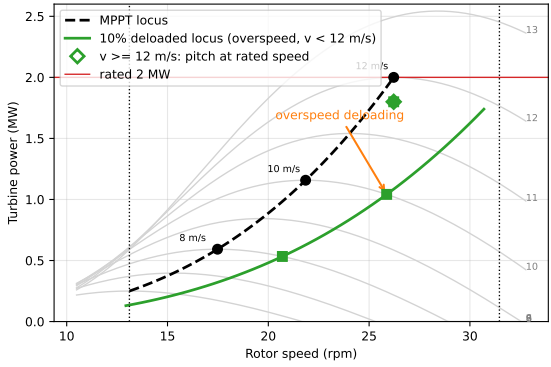
- Converter-interfaced wind: no natural inertia
- GFL (PLL) control weakens at low SCR
- About 54 GW wind installed by late 2025
- GRID-INDIA GFM paper (Feb 2026): BESS and solar focus, little on wind
- Ember: 68 events (Jan 2022 to Jul 2025) with > 1 GW RE lost in one incident; PLL instability, Rajasthan
- CEA amendment applies from 1 April 2027

Link to course theory



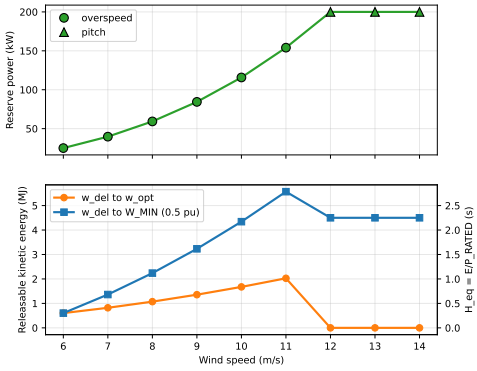
Betz 16/27; $C_p(\lambda, \beta)$; MPPT on the C_p-optimal line, $P = K_{opt}\omega^3$

Key idea: overspeed deloading



Operate right of the optimum: slowing the rotor raises P_{aero}, so no secondary dip

Reserve and kinetic energy



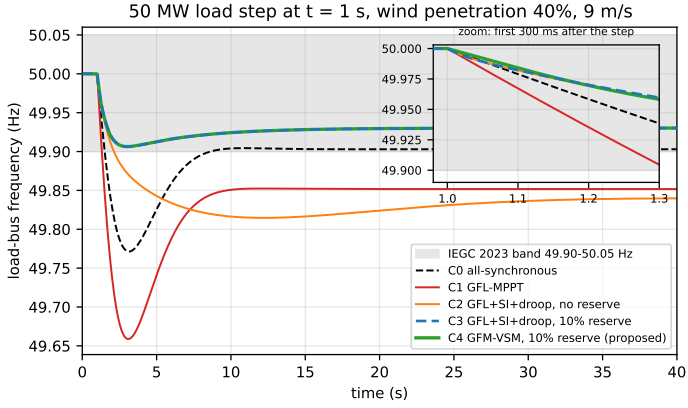
Deload to 0.9 $C_{p,max}$; reserve and kinetic energy released together

Test system and cases

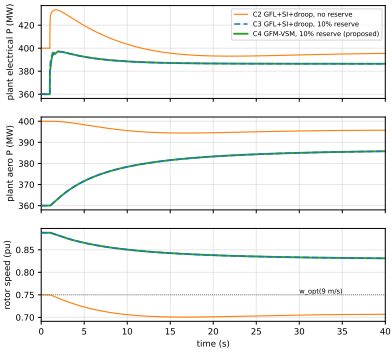
Equivalent SG fleet, reheat governor; DC power flow, $X_{13} = 0.25$, $X_{23} = 0.20$ pu (1000 MVA); 40% wind; 50 MW load step.

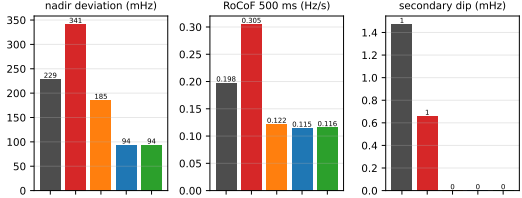
Case	Description
C0	Placeholder: baseline
C1	Placeholder
C2	Placeholder
C3	Placeholder
C4	Placeholder

Frequency result

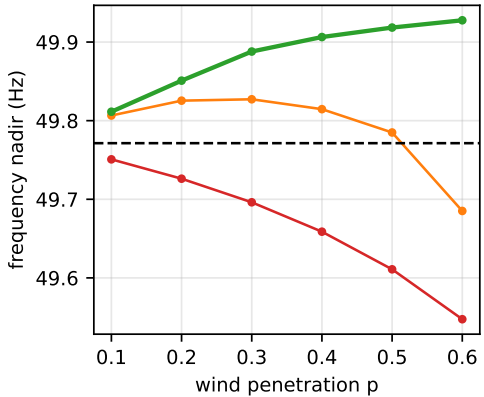


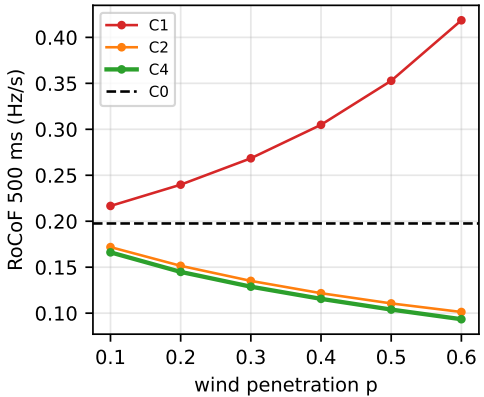
Why GFM wins


C2 GFL+SI+droop, no reserve
C3 GFL+SI+droop, 10% reserve
C4 GFM-VSM, 10% reserve (proposed)
plant electrical P (MW)
plant aero P (MW)
rotor speed (pu)
w_opt(9 m/s)
time (s)

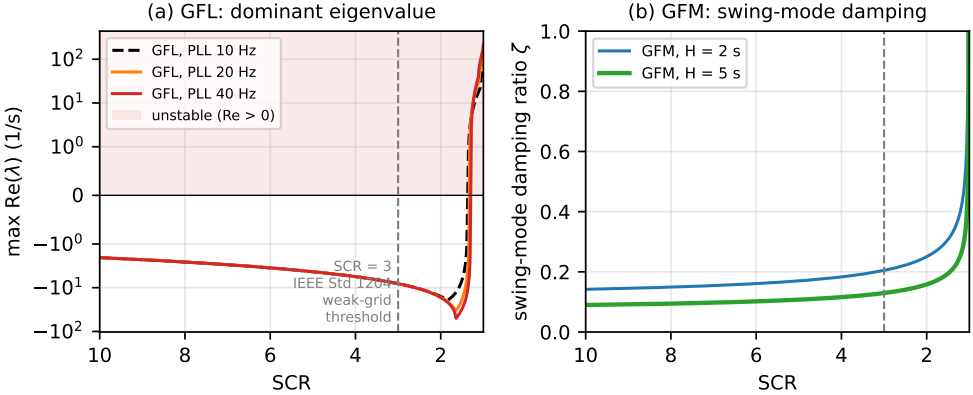

nadir deviation (mHz)
229
341
185
94
94
RoCoF 500 ms (Hz/s)
0.198
0.305
0.122
0.115
0.116
secondary dip (mHz)
1
1
0
0
0
C0 C1 C2 C3 C4

Penetration sweep





Weak-grid duality



Verification checks

Every model was checked by scripted gates against analytic results or invariants (tolerances are those coded in the simulation scripts). All 16 gates of Tasks A, B and C pass (Table 1); none failed and none is missing.

Table: Verification gates of Tasks A, B and C.

Gate	What it checks	Value	S
GA1	$C_{p,max}$ is below the Betz limit	$0.4800 < 16/27 = 0.5926$	P
GA2	λ_{opt} and $C_{p,max}$ lie in the band expected for this rotor (7.5 to 8.7; 0.47 to 0.49)	$\lambda_{opt} = 8.100$, $C_{p,max} = 0.4800$	P
GA3	Rated power at rated wind speed (tolerance 1 percent)	$P_{avail}(12\ \mathrm{m/s}) = 2000$ kW	P
GA4	P_{aero} re-evaluated at each deloaded operating point reproduces P_{del} (6 to 14 m/s)	maximum relative error 6.5×10^{-14} (limit 10^{-3})	P
GA5	Deloaded speed is on the overspeed side, $\omega_{del} > \omega_{opt}$, at every overspeed wind speed	6 to 11 m/s, e.g. 23.28 against 19.67 rpm at 9 m/s	P

Next steps

- Step 2: detailed PMSG dq model, machine and grid converters, DC link
- Dual-port GFM (Lyu and Gross); current limiting, fault ride-through
- Step 3: IEEE 9-bus, turbulence, Indian grid-code comparison, final report